

5. Modulation of autoISF aggressiveness.

V 4.4

Please note that with autoISF you are in an early-dev. environment, where the user interface is **not optimized for safety** of users who stray away from intended ways to use. Good safety features exist, but these are only as good as the development-oriented user understands and implements them. This is not a medical product. Refer to disclaimer in [section 0/readme.md](#)



5.1 Automatic modulation of loop aggressiveness

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Summary: Your personal FCL cockpit

[Available \(related\) case studies:](#)

Case study 5.2: Sweet snack.

Case study 5.3: Compression low

Once the initial tuning according to [section 4](#), is done, you are ready to use autoISF for your fully automated *meal management*.

Challenges beyond managing main meals

There are up to four *other major challenges* you might have to manage:

1. Secure the technical pre-requisites for your FCL remain a given (see [section 1](#)):
 - Prevent, recognize and manage (partial) occlusions (=>frequent cannula changes)
 - Secure “meaningfulness” of 100% all your CGM values
 - Secure 24/7 BT connectivity of all your components

2. Deal with times when insulin given, after meal detection, must be restricted, e.g.:

- a snack should not be “treated” as a meal
- meal before exercise should not be treated “all the way” towards high iob.

3. Deal with times when the loop should be set "less aggressive" as a precaution, e.g.

- during nights
- in an exercise context

4. Deal with times when the loop should be set "more aggressive" e.g.

- at sickness
- at tempting hotel breakfast buffet, or other “diet sin” + in-activity

How big the remaining challenge really is, depends very much on your targeted %TIR and:

- a) on your **individual lifestyle**: => is your loop (automatically, [section 4.1](#), or via a needed intervention, [section 4.2](#)) prepared to respond with required different aggressiveness?
- b) on **quality of your tuning**: => will your loop frequently “drift outside of the desired corridor”? Will it automatically recognize this, and pull back into it? Can you recognize it, and pull back earlier?

Hands-off FCL – an elusive goal?

To run the FCL “hands-off” around the clock, you now must analyze **your** data also in the times *outside* the meal blocks, and seek solutions to problems (if any).

With a technically well-functioning system, *moderate meals, moderate exercise*, moderate %TIR expectations and a bit of mindfulness it should be possible to go into **Full Closed Loop 24/7**, after working through, and observing, [sections 1-4](#). See [case study 4.3](#).

- In [section 5.1](#) we explore avenues towards **fully automated** management of *some other* “disturbances” that, in daily life, also will not require any user intervention.

Often it will be **your** choice whether you want to bother with researching in **your** data for, and defining, dedicated (“personalized”) Automations (see also [section 5.1.6](#)).

Or whether you accept instances where you do manual steps:

- In [section 5.2](#) (and later, related to exercise, in [section 6.](#)) we will look at solutions that involve an easy **user interaction** like a data entry, or a button push.

Avenues for temporary (automatic or manual) modulation of the FCL aggressiveness

1. temporary shut-off SMBs (via odd -numbered target)
2. temporary change bgAccel_ISF-weight
3. temporary change iobTH_percent
4. temporary change the set %profile
5. temporary set different bg target (especially in connection with exercise mode)t

All of these are also easy accessible via Automations in AAPS

This means: If, in **your** data, you find sets of conditions where any of the actions 1.-5. could help, you can make this aspect integrated part of **your** hands-off FCL!

5.1 Fully automatic modulation of FCL aggressiveness

The following subchapters describe set-ups you may want to use for allowing **completely hands-off FCL** in as many daily situations as possible (and potentially all the time, as in [case study 4.3](#))

Note that in Automations you can go beyond above mentioned 5 avenues, and come up with a nearly surgical definition as to from which iob on, and many other criteria, some temporary modulation of aggressiveness shall happen.

5.1.1 All autoISF ISF adaptations switched off outside of meal-time windows

If, aside from having to bolus for meals, your hybrid closed loop was running pretty well *without* other interventions from your side, you could continue to run in that mode, and just focus your new autoISF FCL on management of meals.

In your initial transitioning phase, this approach makes a lot of sense, and even by focusing autoISF on just a sub-set of meals, like only dinners.

Also in the long run this avenue is taken by many FCL users for the night times, “hanging on” to their well performing hybrid closed loop with standard oref(1) SMB+UAM

To install this mode, you define Automations

- that set meal time windows in which “Enable ISF adaptation by glucose behavior” (autoISF) is turned on in AAPS preferences/OpenAPS SMB
- or: that turn *all* autoISF's ISF modulations (or just *bgAccel_ISF*) off in time windows in which surely no meal occurs.

For instance, you can go *for all nights* back into your Hybrid Closed Loop, as you had before.

Your temp. “autoISF shut-down” (exiting autoISF FCL = shutting off “Enable ISF adaptation by glucose behavior” in AAPS Preferences / SMB) is meant to prevent problems from the loop *over-reacting* to bumps in the glucose curve in times of day (night), when standard oref(1) performance is sufficient.

A very good alternative to defining “meal-time windows”, or to resorting to night-time Hybrid Closed Loop, is letting the autoISF FCL run 24/7, and “taming” the FCL via a night time SMB shut-off (see next [section 5.1.2](#)).

Off topic: **Other “early dev” AAPS variants** (see [section 13.3](#)), do require working with meal-time windows. The window is either set by time of day in the settings, or it always must be „set“ by the user. Trigger to set a meal time window could be a pre-bolus given by the user, a carb entry made, an EatingSoonTT set, or a meal-announcement-button pushed (none of these things are required in autoISF FCL). Outside of these time windows, these loops then run with less aggressive SMBs, just like oref(1) SMB+UAM in AAPS Master, with or without (modified versions of) dynamicISF. So, outside of the meal time windows, you would be in “your normal” hybrid closed loop. The term **Meal Announcement** (MA) is often used to label this closed looping mode. It is not really FCL, but an advance over traditional HCL.

5.1.2 Odd-numbered **profile targets**, to block SMBs

An alternative route of preventing the FCL loop from over-reacting to bumps in the glucose curve would be to make use of the option to temporarily shut down SMBs

Ensure the even/odd logic in the settings is toggled on in Preferences> openAPS SMB> autoISF settings> smb delivery settings>: "Enable alternative activation of SMB depending on bg target": ON.

154

In time blocks with an odd-numbered profile target you can prevent any SMBs being given by your loop. The (unchanged) aggressive settings then can only translate within the limits set by %TBR possible.

158

This will very much slow down any more insulin being given, and is an excellent solution for night times, especially if you occasionally experience compression lows.

161

Alternatively, you could use the new included options for Automation Conditions and temporarily tune your bgAccel_ISF_weight much lower (see [section 5.1.4](#)).

164

The same situation can be achieved if you generally operate with a mild bgAccel_ISF, and make your autoISF only really aggressive for meal-time slots (if you have similar enough times every day, or also can "employ" geo-fencing in your Automation (or middleware, in iAPS) conditions).

In these cases you would not need to have night profiles that disable SMBs: - Which is the better way would depend on a lot of personal factors relating to how high-carb the diet is, regularity of meals, snacking habit, CGM quality and incidence of compression lows, and probably more. - I would try both routes, or, as this is fairly complex to tune, just one, and stick with what is working good enough.

Yet another alternative was already presented ([section 5.1.1](#)) = to go into hybrid closed loop for the night.

That is possible, with SMBs available (without them getting boosted via autoISF), and, for a long time, was the author's favored solution for the nights.

This solution is evidently similar to the prior discussed one, of having a mildly tuned autoISF 24/7, boosted to high aggressiveness only in meal-time slots.

179

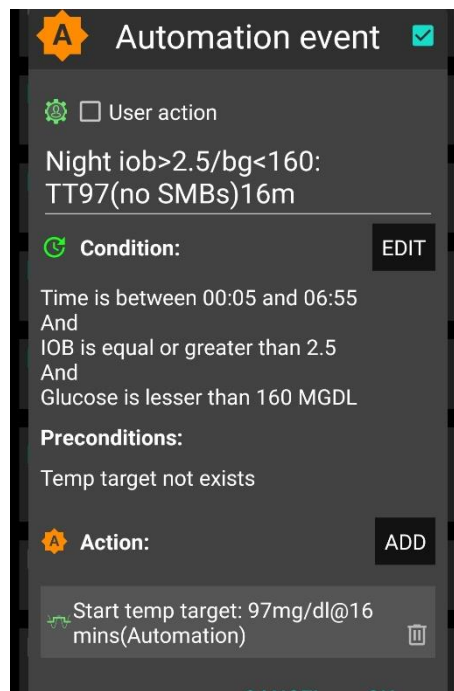
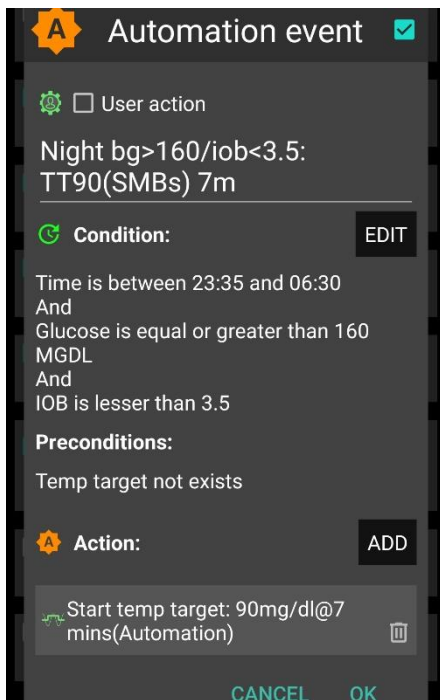
[Adjunct Automations to allow a few SMBs, in nights with odd profile target SMB shut-out](#)

181

My **favorite** builds on the **method** of this section (5.1.2, odd **profile** target provides SMB shut-out at night), but then allowing some, automatically triggered SMBs, when needed:

184

In case you occasionally do have nights that would benefit from a couple of SMBs (to treat temp. highs from a late fatty pizza, raclette and such): Define suitable Automations like the two „night“ ones in this *example*:



188
189

190 Caution: Never underestimate the „trickiness“ of getting your Automations „right“.

191

192 With your thought-out Automations in place, night data need to be analyzed to see

- 193 • whether the bg and iob limits, as defined in the given example, work sensibly four your data
- 194 pattern
- 195 • whether the TT duration is chosen appropriately
- 196 • how swapping the sequence in which the automations appear in the Automation list would
- 197 lead to different SMB impacts.

198

199 5.1.3 Odd-numbered **temp. targets (TT)** set via Automation, to block SMBs

200

201 A widely used Action that strongly modifies how fast your FCL can add more iob is setting an **odd-**
202 numbered **temp. glucose target** which makes the loop operate without giving any SMBs (%TBR
203 modulation only).

204 Ensure the even/odd logic in the settings is toggled on in Preferences> openAPS SMB>

205 autoISF settings> smb delivery settings>: "Enable alternative activation of SMB depending
206 on bg target": ON.

207

208 Then, from patterns you find in *your* data, at times where you want your loop act differently, you

209 need to carve out Conditions that describe the respective situations (and either *for how long* it

210 typically lasts, or at which *other* Conditions you want your loop get back to default FCL operation).

211

212 An odd TT is often set for an *anti-hypo* snack or *sports* snack. In both instances, you do not want
213 SMBs to quickly counter-act.

214
215 Note that in case of sweet “fun” snacks, this is entirely different -> [section, 5.2.1](#) or for
216 regular snacks (*e.g. at school break*) see next [section 5.1.4](#)

217
218 **5.1.4 Automatic differentiation of FCL aggressiveness using Automations (or middleware)**

219
220 **Personalized Automations** tailor the loop exactly to **your** data so **fully automated** handling of
221 situations with **different aggressiveness** of the loop can be made.

222
223 Automations are an integrated and very easy-to-use feature in AAPS.
224 (The i-Phone platforms Trio or iAPS lack this feature. However, so-called **middleware** has been
225 developed as add-in to your code, see: <https://github.com/macconnellk/RoboSurfer/tree/main>)
226 .

227 From, autoISF 3.0 onwards, also the following parameters are provided as Condition and/or as
228 Action for defining YOUR Automations:

- 229 • Enable ISF adaptations by glucose behavior => Allows temp. ON/OFF for the key ISF
230 modulation parts of autoISF (and, as a result, will usually decrease loop aggressiveness)
- 231 • Trigger/set iobTH percent => Keeps default aggressiveness, but only until a iob threshold
232 (that your Automation modifies) is surpassed (which is when any further SMBs will be
233 blocked blocked)
- 234 • Trigger/set bgAccel_ISF_weight => Modifies the aggressiveness of just the acceleration
235 component

236 To set up suitable Automations, you first must **analyze patterns** you find **in your data**, at times (or
237 geo-location, or bg and iob patterns that point to a problem ...) **where you want your loop act**
238 **differently**, to carve out Conditions that describe the respective situations (and either for how long
239 it typically lasts, or at which *other* Conditions you want your loop get back to default FCL
240 operation).

241
242 A variant of this mode is to define several windows in which autoISF aggressiveness
243 (bgAccel_ISF_weight) and/or iobTH are automatically set differently

- 244 • for **different meal time slots** of your day –

245 (*Breakfast at home, school lunches, school intermission snacks, dinners at home* could for
246 example all deserve special settings regarding ISF_weights and iobTH).

247 Note: Circadian differences in insulin sensitivity between meal times are included via your
248 ISF profile and should not be a reason for different _weights needed between meals!

249 • or even for a geo-location etc –

250 (*School lunches, or mother-in-law visits, would be examples*).

251 An example for this was given in section 3 already:

Here is an example set of automations to alternate between two values of iobTH:

I use two different values of *iob_threshold_percent* during a normal day. It is 40% for lunch time and 60% for dinner time. I have these two rules to switch by time of day and only if the current value equals the value from the earlier shift. Any other value is treated as a manual override for special occasions until I manually set it to its regular value. The time windows for switching are long enough to catch an opportunity to be processed and do not need to be actioned half a day each.

252

253 Unless your meals differ vastly in size and in fast carb content **all this may not be needed**.

254 • Note: Only the main two parameters (bgAccel_ISF_weight for “initial aggressiveness”, and iobTH_percent for
255 “where SMBs stop”) are available in Automations. So, finding your parameter sets *for each of the time slots*, will
256 not be trivial. => **Spending more effort to set the .._weights so they accommodate just one, broader**
257 **spectrum (section 4.) should be the first, and standard, approach.**

258 • An intermediate (maybe only temporary) approach could be to use a **profile switch** (for low carb meal, or
259 eating half, setting %profile to 60% for instance, and only for the brief, less than an hour, initial meal period).
260 See [section 5.2.2](#)

261 Still, personalized Automations might help ease your initial job of setting the various ISF_weights,
262 and of finding a best-suitable iob_theshold_percent that would work “always”.

263

264 **Note:** Setting a different iobTH% or bgAccel_ISF_weight can probably not be done with a duration
265 attached. Then you **must** define a suitable **additional Automation** (example see next page) that
266 must be active in tandem, to **restore the values you had set in /Preferences for your iobTH%**
267 **or bgAccel-ISF_weight.**

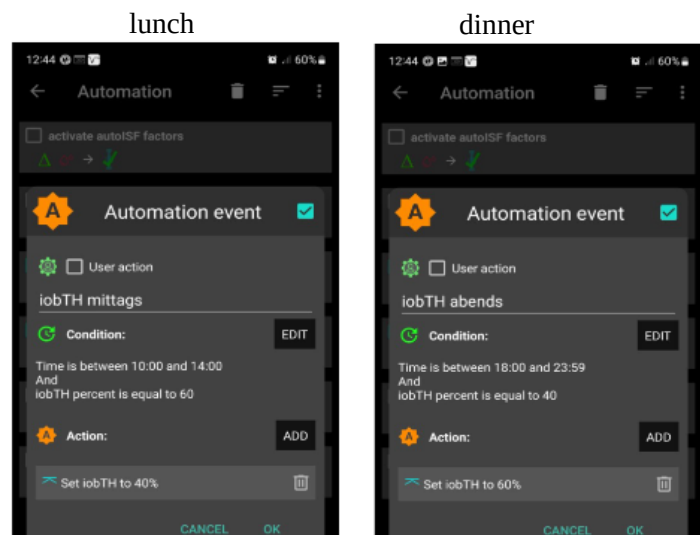
268

269 Else, once your Automation set in, it will *forever* shift these important parameter settings!

270

271

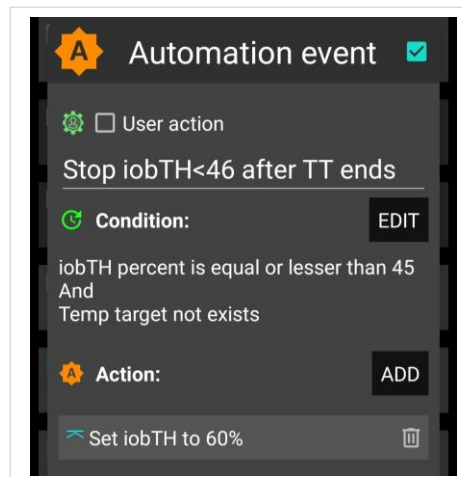
272



273 If for instance you have several Automations that, in
274 combination with a set elevated TT also set a lower iobTH:
275 Don't be fooled, the duration only applies to the TT.

276
277 You need an extra Automation for all of them.

278 I picked out the highest of the altered iobTH values that these
279 Automations can set (*in my case, 45% or lower*), and then I can
280 automatically restore *my default desired 60%*, via *this one* - ->



282
283 **Caution:** Resist the temptation to use too many Automations. Notably, do not include Automations
284 that would duplicate (and in effect delay) the job that autoISF tries to do:

285 Automations: Relative **disadvantages** to/when using autoISF ([Discord 26.8.2025/ chiakii/ ga-zelle](https://discord.com/channels/2682025/2682025/2682025)):

286 “..I've made an **automation** that sets a low TT on high bg deltas, but when it gets triggered, the SMB I get is calculated
287 with the "before TT" weights/values, and only the next tick (next CGM value) will take the low TT into account.
288 Given that we want to front load the SMBs as much as possible, that's **5 minutes lost** with a high delta, which isn't
289 ideal”

290 ANSWER: A new bg reading triggers a new loop and checking the automation triggers. Therefore it comes too late.
291 Your scenario reminds me of cases with too **many automations** that **interfere with the** (autoISF) **built in control**
292 **logic!** Note that if you observe large delta, then inevitably there was a high acceleration **before**. In terms of front
293 loading that **acceleration** could have done its job when you use **autoISF**. Even a stronger **pp_ISF** weight would
294 probably help more.

296 5.1.5 Automatic differentiation of FCL aggressiveness using “Automation States”

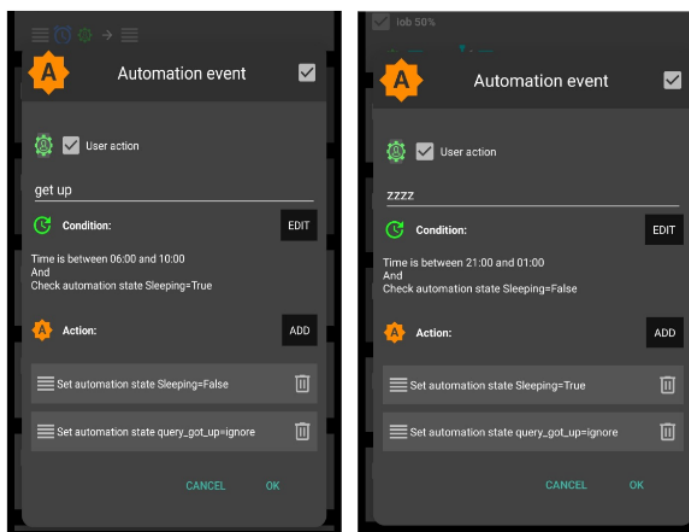
297
298 This is a prototype implementation in autoISF 3.1.0 that allows further refinement of automating with autoISF
299 parameters.

First real use example for disabling inactivity monitoring during sleep

300 An application
301 example is given in
302 the developer's
303 autoISF 3.1.0
304 Quick Guide,
305 section “The new
306 Automation of
307 States”:

308 Source:
<https://github.com/ga-zelle/autoISF>

My sleep times are very irregular and I do not want to adapt those hours in the preferences every night. So I created a *state* Sleeping with its two obvious *values*. In addition I created two user action buttons which swap the values for going to bed and getting up, respectively. They trigger visibility in the morning or evening based on likely time windows and *state values*. The real trick now is to interact with those *states* inside the AAPS code rather than just by automations. This offers more complex behaviour. The slight downside is that in this case you have to stick to the names of *state* and *values* shown here. This will take precedence over the sleep hours and its enabling switch. If you do not want this new feature just do not create the *state* Sleeping – or misspell/rename it for your own use.



309 5.1.6 Automatic adjustment of FCL aggressiveness via the Activity Monitor

310

311 With the autoISF variant of AAPS you can make use of your smartphone's **stepcounter** and use it
312 to fully automatically adjust insulin sensitivity ratio to **activity level in the past** minutes to **one**
313 **hour** time frame.

314

315 This feature comes with yet another little tuning opportunity, in which you study your body's
316 response to light exercise (like walking) or to not moving at all (like desk, couch), and select
317 appropriate settings (in AAPS Preferences/OpenAPS SMB/Activity modifies sensitivity/ -> set **two**
318 **scaling factors**).

319 This **will automatically adjust insulin delivery** (basal, ISF, and iobTH; see 1st screen of AAPS
320 SMB tab (example in [section 5.4.5](#))), to suit activity state of the past minutes (up to 1 hour).

321

322 This autoISF feature (new since V.3.0) is much quicker responding than Autosens or dynamicISF
323 to adjust insulin sensitivity to your current „lifestyle state“ (if it is largely (in-)activity related, which
324 often is the case).

325 For loopers who do not have huge variations in exercise levels (for which they would need the
326 exercise mode) in their everyday lives, the Activity Monitor might fairly much close the gap towards
327 being able to do a 24/7 hands-off FCL.

328

329 The Activity monitor is described in more detail on page 9 of the autoISF Quick guide
330 (<https://github.com/ga-zelle/autoISF>) and in [section 6.6](#)

331

332 While the Activity Monitor takes automatic care of *light* deviations from an average activity level,
333 exercise enthusiasts, or heavy workers, should make use of the “*heavier tools*” (which
334 automatically supersede (shut off) Activity Monitor: *Exercise mode*, discussed in [section 6.](#))

335

336 5.1.7 Pro/con completely hands-off Full Closed Loop

337

338 To stay **24/7** in a **completely „hands-off“ FCL** can be a realistic goal with autoISF, if besides
339 meals also some special challenges, as discussed in this [section 5.1](#), were analyzed and could be
340 addressed. Example see in [Case study 5.3](#).

341

342 Clearly it depends very much on your lifestyle, and how interested, willing, and capable you are to
343 recognize, deal with, (and in the future avoid?) situations that get you outside of your desired %TIR
344 on occasion.

345 So, this is also about what %TIR you are aiming at, and can accept, as it averages out for
346 the week, for instance.

347 Everybody must weigh for her/himself

- 348 • how much **upfront effort** to put into the setting up process, for getting it all 100% automatic
- 349 • **or** whether to take an **easier start, with a couple of situations left to take care of when**
- 350 **and as they arise in daily life**

351

352 Even if a principal capability for a fully automatic “hands-off” running FCL is given, this still
353 means that

- 354 • the user should be knowledgeable about what exactly is going on, and
- 355 • have a principal capability to „nudge“, or even to completely take over, in a manual mode.

356 In the sections that immediately follow, we present the options to nudge or temporarily take over
357 from the AAPS home screen which will be serving as your **FCL cockpit**:

358

359 [Section 5.2](#) describes how you can use available “buttons” from your AAPS home screen to
360 **manually** “nudge” aggressiveness of your FCL.

361

362 5.2 Modulating aggressiveness **manually**, from your DIY-FCL-Cockpit

363

364 Keep in mind:

- 365 1) The goal should be to **interfere with the loop as little as possible**
- 366 2) Avoid technical failures and BlueTooth disconnections lasting many minutes (e.g. by leaving
367 your phone in another room for too long).

368 The required insulin would still be supplied *after* BT reconnects. However, the delay can be much more
369 of an issue in FCL than it had been in HCL, where you pre-bolussed.

370

371 In the preceding [sections 4. and 5.1](#) we learned that, under certain conditions, your FCL can run
372 **fully automatically**, without any user interaction.

373

374 However: *Other* **disturbances** might come up, that:

- 375 • are not noticeable in-time, or foreseeable, by the loop (*e.g. your plan to start exercise in an*
376 *hour or two*), but **that influence sensitivity dramatically**, and therefore require temporary
377 modified settings in order to remain in-range, and/or
- 378 • **require** a different “starting point” regarding iob and bg, which translates into **a different**
379 **iobTH** that should **temporarily** be set much lower (*in case of exercise*) or noticeably higher
380 (*e.g. with very fast absorbing carbs in a sweet snack “sin”*) .

381

382 *In these scenarios, you must find an easy way to*

- 383 • manually tweak a setting or two, to temporarily adjust the aggressiveness, or:
- 384 • call up a pre-programmed routine for automatic management, with adjusted
- 385 aggressiveness
- 386 • There may also arise a desire to just exit the FCL mode, and “be your own captain” for
- 387 mastering a special situation.

388

389 5.2.1 Status recognition

390

391 For peace of mind, to learn, and to stay informed (especially so in your initial tuning phase, or
392 when your glucose curve goes in unexpected ways) you must be able to find the key parameters
393 that frame and drive the recent and upcoming loop decisions.

394

395 All this is facilitated within seconds right from the AAPS home screen, serving as a **FCL cockpit**

396

397 Before considering any manual interventions into the ongoing FCL, you should be aware of

- 398 • the prediction curves (bg, and insulin activity), and
- 399 • what the current mode of action is (refer to [section 5.3](#)).

400

401 On that basis you might be able to “nudge” your loop in order to adjust to the disturbance that you
402 see coming up:

403

404 5.2.2 Manual interventions from the (DIY-) FCL cockpit

405

406 5.2.2.1 Temporary tuning of FCL aggressiveness via the top button row on AAPS Home screen 407 (temp. %profile; exercise; TT settings)

408

409 %profile set

410

411 The set **% profile** multiplies with both, the ISF resulting from autoISF, and also with the default
412 iobTH you have set, so both are nicely modulated in a linear way with the % temporarily chosen

413

414 Just taking profile e.g. to 110% for an afternoon might be an easy way to explore whether
415 you might benefit from 10% more “aggressiveness” in your core settings for lunches

416

417 Make sure, though, that your safety limits ([section 2](#)) are set wide enough so they will not
418 cut the extra 10% away!

419 Temporary bg target (TT)

420

421 A **lowered** (relative to profile glucose target) temporary **bg target (TT)** signals lowered sensitivity
422 (more insulin need), e.g. as an “EatingSoonTT”.

423

424 An **elevated** TT (e.g. used with exercise) increases sensitivity and hence works in the direction of
425 reducing insulin given by the loop.

426

427 **Caution:** In preferences/SMB..., make sure you set “High TT raises...” and “Low TT
428 lowers sensitivity”.

429 However, advice is NOT to activate the other two options there, that would automatically change TT depending
430 on detected sensitivity. One reason: With our important even/odd TT feature (regarding SMBs, see next
431 headline), we would hate to see TT randomly set to even or odd values.

432

433 Exercise button

434

435 Moreover, the **exercise button** (top center on your AAPS home screen) can be activated (turns
436 yellow, then). This can ***) further boost** how your set TT elevates the resulting ISF, and sharply
437 lowers iobTH, as often desired for sports. ***) see below in [section 5.3.2](#), and more in [section 6](#)**.

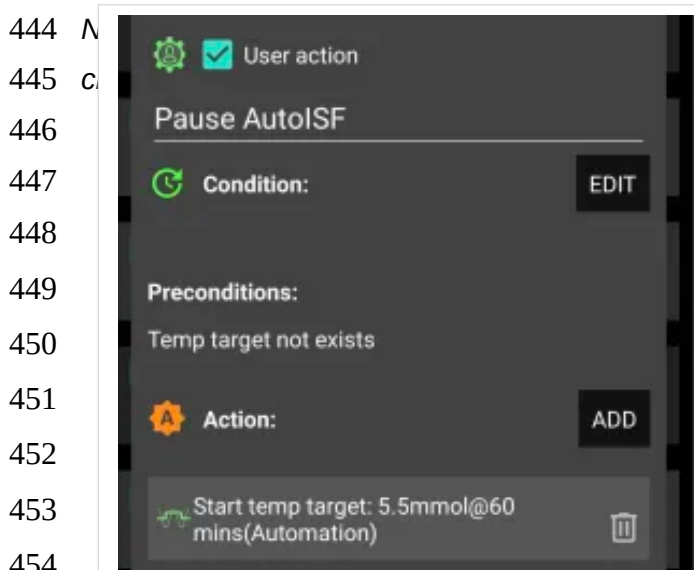
438

439 Even or odd numbered temp. bg target (TT) (of similar level)

440

441 As we learned in [section 5.1.3](#) already, switching between an even or odd bg target can be used to
442 strongly modify loop aggressiveness by switching SMBs on/off temporarily.

443



444

445

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456

It is easy to use the **top right “TT” button** in the AAPS screen to switch from an even target to an odd numbered one, in order to shut off SMBs for a time you can set.

<- An even easier to operate alternative is setting up this **User Action Automation** called “Pause autoISF” (suggested by Slip, Discord Dec.2024).

(Actually, autoISF keeps running, but it *seems paused* because it can’t translate big calculated insulin requirements into SMBs)

457 5.2.2.2 Making temporary changes in settings made in AAPS/preferences/Open APS SMB

458

459 **Caution:** Trouble with some of these is, not to forget to stop and set them back, manually, too.

460

461 Going into AAPS/preferences/Open APS SMB allows to:

- 462 - set milder or stronger ..._ISF_weights
- 463 - set different iob_threshold_percent (or iobMAX)
- 464 - elevate or lower the SMB_delivery_ratio
- 465 - limit or expand max. allowed SMB size
- 466 - change the the even <-> odd logic for SMB on/off

467 Doing temporary changes *in AAPS/preferences* **should be the exception because**

- 468 - they require multiple steps, including entering a password
- 469 - you will often forget to set everything *back to original* settings, a couple of hours (or already
- 470 minutes) later.

471

472 5.2.2.3 Triggered Automations: Grey extra DIY cockpit buttons for pre-programmed “responses”.

473

474 In AAPS, you can build a couple of DIY cockpit features via “User Action Automations”, as
475 described below and in [case studies 5.2](#) and [6.2](#):

476

477 **Snack example.**

478

479 Recognizing conditions for **fully automatic** handling by the loop may not be not possible, or come
480 too late for the loop to act on. Examples would be

481

- 482 • *exercise*: Minimum an hour *before* starting exercise, “the loop should know” to be able to
483 lower iob and elevate bg by the time exercise starts. (For more, see [section 6](#).)
- 484
- 485 • *snacks*: High carb snacks, sweets, consuming ice cream or having a sweet drink comes
486 with the problem of even steeper glucose rises, but overall a lesser insulin need, compared
487 to major meals (for which we tuned our FCL according to [section 4](#)).

488

489 This not necessarily implies that snacks *need* different settings than a meal. After all, autoISF
490 was designed to react to all available data, especially to where the developing glucose curve is
491 headed. So, depending on your effort to set parameters for a broad variety of meals (notably:

how well you avoid to invariably bounce fast against your iobTH), you might be able to accommodate low carb, snack, and major meals with *one* set of settings.

In FCL autoISF, this is a bit more difficult than in HCL autoISF applications, because FCL involves revving up iob supply (largely via big bgAccel_ISF-weights) sometimes too much, to be balanced by just a snack getting absorbed.

Caution: In case a snack *did* trigger a “full meal response”:

- (1) You probably must continue snacking to prevent a hypo from your initial FCL over-reaction.
- (2) For future days, analyze your data (and snacking habit) to define how to prevent this from happening often.

For increased comfort and safety, you might have to differentiate, and make use of what follows for the *sweet snack* example, [case study 5.2](#).

Note that in the iPhone versions of autoISF (**Trio** and **iAPS**) there are no Automations . Instead you need so-called **Middleware**, like for instance suggested for %sensitivity (profile ISF) adaptation by one user here: <https://discord.com/channels/953929437894803478/1025731124615458848/1238099464531611668>

Tuning aggressiveness

A sweet snack likely benefits from even more aggressive initial FCL performance than set for the meals in your normal spectrum of diets.

Therefore, you could set

- a higher **temp. profile%** and/or
- a temp.elevated **bgAccel_ISF-weight** (see screenshot of my Automation).
- a **low temp. target** (76 mg/dl for instance; this additionally helps maximize the first SMBs that will automatically be triggered at detection of acceleration)..

When first defining and testing this Automation, also check:

- that the safety limits as discussed in [section 2](#) will not block the intended elevated aggressiveness
- SMBs will not get outrageously big, and iobTH sometimes exceeded by too much.

Note that “the last SMB” is allowed to overshoot the effective iobTH by up to 30%, where it will be cut.

529 Limiting iob

530

531 For “just a snack”, total insulin need will be lower than for a meal.

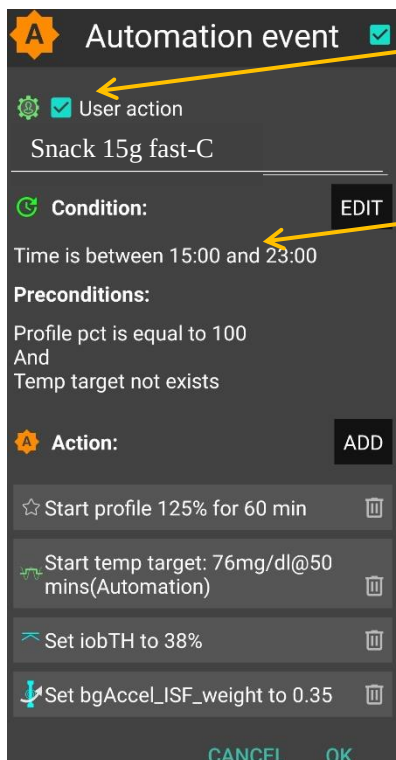
532 If you would just have your sweet drink, and your meal-oriented FCL would “attack”, iob

533 likely would become too high, and a glucose rollercoaster would start, with you needing to

534 consume more =>

535 If you just have a snack, or drink a small glass of juice, you can lower the **iobTH_percent**

536 accordingly.



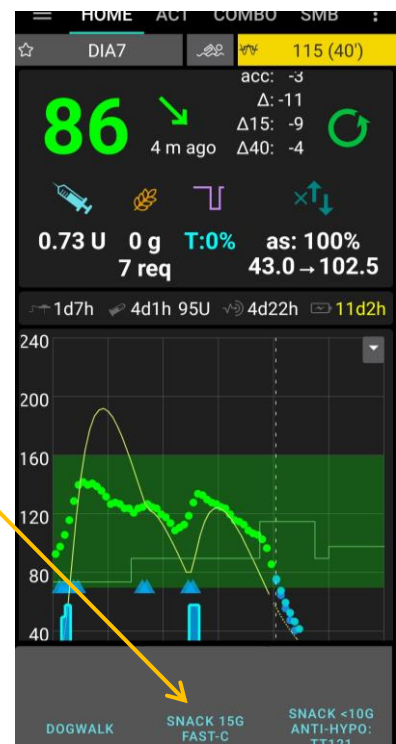
„User action“ is always ticked-on

This will, in the defined time space ..

..offer the “DIY cockpit” button..

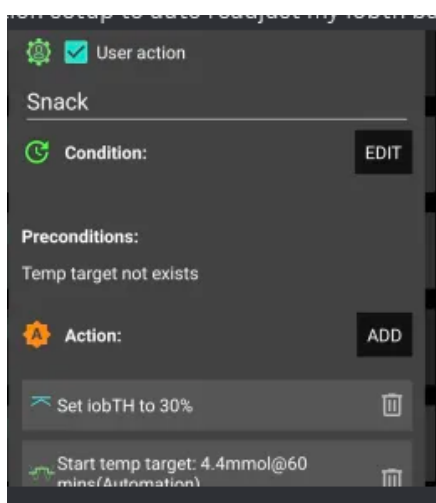
..which I must press any time (~90...30 minutes) before my snack.

Note: Make sure that iobTH and bgAccel_ISF_weight revert to default afterwards



537

538



<.- A simplified way was proposed in Discord (Slip, Dec.2024)

Again, note: Make sure that iobTH% reverts to default afterwards, ideally automatically:

- see Automation event with Action “set iobTH (to default)” (further below)

539

540

541 So, you as part of your setting-up project, you need to research your snack habits (if any),

542 and over time find out which settings in the snack-related Automations work well. Then, **in**

543 **everyday life, you just press the related button in your cockpit** (which is not time
544 critical at all, except it should be clicked **latest** a couple of minutes *after* you took the drink
545 or snack, when acceleration of your bg curve sets in.)

546

547 If you consume more, and also eat something with your sweet drink, this will more
548 resemble a full meal (... probably with unusual amounts of fast carbs => it is good to use
549 initially the more aggressive ISF weights as in your snack Automation).

550

551 **Caution:** Pressing your snack button *a second time* would **not** help because the lowered
552 iobTH does not allow iob going high enough. So you are better off just letting your *normal*
553 FCL meal routine run, after your snack mode expired.

554

555 Other options (when you just can't stop snacking) would require a manual modulation
556 regarding %profile and/or bgAccel_ISF, but keeping the full default set iobTH_percent, or even
557 elevating it (refer to [section 5.2.3](#)).

558 If that happens often, define for yourself an *extra* User action Automation for a *bigger* snack (=
559 another grey DIY cockpit button).

560

561 **Caution:** Setting a different iobTH or bgAccel_ISF _weight can not be done with a duration
562 attached.

563

564 Hence you **must** define a suitable **additional**
565 **Automation, that** must be active in tandem, and
566 **restores the iobTH or bgAccel-ISF_weight** in
567 AAPS/Preferences.

568

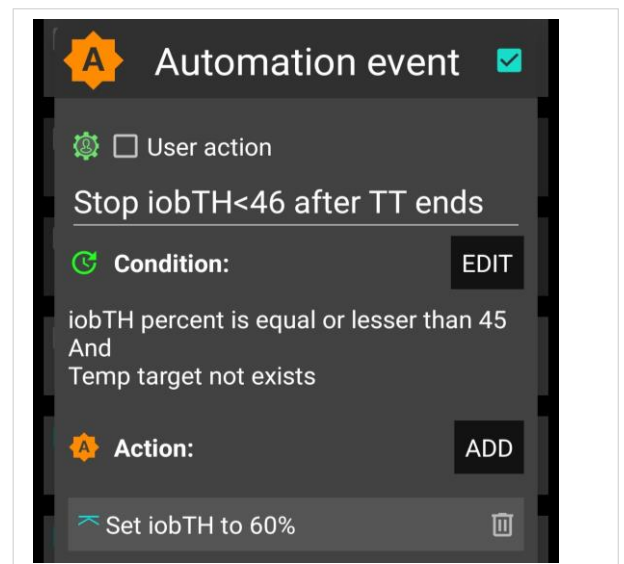
569 Else, once your Automation set in, it will *forever* shift
570 these important parameter settings!

571

572 Tandem Automation example, for restoring settings in
573 /preferences : ----->

574

575 If for instance you have several Automations that, in combination with a set elevated TT also set a lower
576 iobTH: Don't be fooled, the duration only applies to the TT. You need an extra Automation for all of them.
577 In my Tandem Automation example given above, I picked out the *highest* of the altered iobTH values that my
578 iobTH-reducing Automations can set (45_percent), and then I can automatically restore my default desired
579 60% via this one Automation (see screenshot given above)



580 In the related Automation, just keep the “User action” box clicked at all times, and define in the
581 Conditions when you want to see that button available for cockpit use (see screenshot above) =>
582 you will see that button offered.

583

584 [Installing your DIY cockpit buttons](#)

585

586 Besides snacks, also any other **recurring special situations can be addressed via a DIY**
587 **cockpit button, and receive different aggressiveness** up to a suitable iobTH level.

588 Over time you can have a big number of User action Automations. Best keep some “shelved”
589 (clicked in-active, top left box) in your long list of potential Automations. Even when active, they
590 only show in your cockpit (bottom grey field of your AAPS home screen) *in the time slot you*
591 *assigned* as potentially relevant (and if all other conditions are met).

592

593 [Summary](#)

594

595 In case you do have a snack habit and ...

- 596 • can not find settings, as in [section 4](#). defined for your meals, that also suit your snacks
- 597 • can not pin a time slot or other Condition to it for programming an Automation response
- 598 as in [section 5.1.4](#)

599 .. then you minimum need a “snack announcement” for which the extra button in your DIY cockpit
600 provides a time-uncritical 1-button-push solution.

601

602 This could be a good solution for kids in kindergarten, too. Make sure caregivers
603 understand to **use it *only once*** for a snack! **Using the snack button several times in a**
604 **row would limit iobTH at a too-low level!** Note that continued snacking would require iob
605 as for a meal, and that is what the FCL loop takes care of, fully automatically.

606

607

608 [5.2.3 Temporarily exiting the FCL](#)

609

610 The “last resort” alternative always is to **temporarily** leave the FCL mode, and handle any
611 disturbance “the traditional way” in **hybrid closed loop**.

612

613 Do not forget that, before any meal starts, giving a bolus will then be necessary again.

614

615

616

617 "Enable ISF adaptation.." OFF"

618

619 This switches *all* automatic aggressive adaptations of ISF to the bg curve OFF that autoISF
620 provides.

621

622 To exit the FCL mode, go to AAPS/preferences/*put in your password *)*/OpenAPS SMB/scroll down
623 to autoISF settings and switch "**Enable ISF adaptation..**" **OFF**"

624 (or, alternatively, set bgAccel_ISF_weight to zero).

625 **) please set a (very short) password, just to avoid accidental clicks when scrolling*

626

627 Set bgAccel_ISF_weight temp. to zero

628

629 If in hybrid closed loop you like e.g. the dura_ISF adaptation still, you alternatively could elect to just
630 set bgAccel_ISF_weight temp. to zero.

631

632 Switching back into autoISF FCL

633

634 **Caution:** Unfortunately, there is no way yet for your full closed loop with ISF adaptations to come
635 automatically back on, after a selected time for instance. So **do not forget to switch your**
636 **autoISF fully back on, later.**

637

638 Alternative: Temp. FCL OFF via a new cockpit button

639

640 As this switching back will often be forgotten, an alternative strategy for FCL pauses would involve
641 a custom "User action" Automation, that can give you a "Pause FCL" grey button on your AAPS
642 Home screen (see end of [section 5.2.2.1](#), above).

643

644 Caution though, there is very limited experience with this brand new feature. Make sure your Automation definition really
645 applies a duration (or other condition) that will automatically terminate all settings changes it made. As we have seen e.g.
646 in [section 5.1.4](#), this is not always the case.

647

648 Recognizing loop state

649

650 To recognize whether autoISF currently runs with **ISF adaptation** or not, please consult the
651 "**ai: %**" **indicator** below the Autosens% **on the AAPS Home screen.**

652

653 The 1st screen of the **SMB tab** is probably the best resource at your fingertips to find out in detail
654 what is currently going on.

655 More see in [section 5.3](#) that follows.

5.3 Recognizing your loop state in the AAPS home screen

5.3.1 Modulated loop aggressiveness via the 3 top buttons in the AAPS home screen

The 3 top buttons (%profile; exercise; TT) of your AAPS Home screen allow temporary modulation of aggressiveness (sensitivity), as may be needed e.g. when

- going into meals (EatingSoonTT => providing lower bg “starting point” and more pos. iob),
- or doing exercise (Exercise mode, limiting basal, iob etc for hypo prevention).

All buttons grey (GGG) indicates: The **loop is running using** un-modulated **profile** parameters

Note that autoISF builds on profile (or on a temp. modulated profile) for further “5-minute” modifications (See calculation cascade for used sensitivity (sens) in the SMB tab. Example see below in [section 5.3.6](#)).

Any **yellow** (y or Y) button indicates you are running with a modified (*elevated*, or **lowered**) **profile sensitivity**.

Actually, the middle button could be yellow without being active, see table below

	Less aggressive loop = more sensitive user needs temp. less insulin	Standard loop using the set profile	More aggressive loop = less sensitive user needs temp. more insulin
% profile	under 100 % (y)	= 100% (G)	above 100% (Y)
Exercise button	ON (y) Note that it will only work in combination with elevated TT	OFF (G) <i>If ON (y), it is <u>not</u> active but ready to automatically become active whenever any TT above target is set (e.g. by any Automation)</i>	OFF (G)
TT	above profile target (y)	= profile bg target (G)	below profile target (Y)
Example	Exercise mode with weakened %profile, yyy		EatingSoonTT, GGY
..used e.g. in :	Case study 6.2 (yyy)		Case study 6.2 (G_Y via Automation) Case study 5.2 (YGY)

[Case study 6.2](#) uses extremes from both ends of this sensitivity modulation spectrum:

- yyy for reducing basal and iob, and getting weaker ISF, during exercise
- GGY for temporary “EatingSoonTT” boost when, in FCL, a high carb meal had

- The even stronger YGY case is contained in [Case study 5.2](#), where for a high carb snack an extra boost, using >100% profile sensitivity, is temp. added

680

681 5.3.2 Color scheme of top cockpit buttons tells kind of closed loop that is running

682 This section is fairly complicated, not fully checked with Devs, and not crucial to understand: Skip
683 and continue at 5.3.3

684

685 3 Buttons (%profile; exercise; TT), each in 2 states, yellow (for modified), else: grey (G)

686 Note that yellow could be less (y) or more (Y) aggressive than standard profile

687 Any yellow setting modulates sensitivity *or attempts to modulate, whenever another required condition*
688 *may become true (e.g. also via an Automation)*

689 Note, though, there are EXCEPTIONS (see below, under GGG, and under Gyy) where sensitivity
690 is auto-adjusted without the user, or an Automation, making a different setting on any of the 3 top
691 buttons. Autosens, or the Activity Monitor, can adjust the %profile sens without the top left button
692 turning yellow or showing that %value. (See sens field in AAPS main screen, or SMB tab!)

693

- GGG is the loop running with set profile.

695 EXCEPTION: If the Activity Monitor or Autosens are running, profile sensitivity could be
696 adjusted to (in)activity despite the %profile button remaining grey (see top of SMB tab text
697 from AAPS main screen).

698 Likewise, Autosens can temp. tweak the set profile, with the button still remaining grey,

699 Overall, 2 exp 3 = **eight principal FCL modes** are possible. They need to be **differentiated**
700 **further**, based on whether the modulation...:

- ...goes into the **more aggressive (Y, higher % and/or lower TT)**, *or...*
- ...the *less aggressive (y, lower % and/or higher TT and evtl exercise)* direction.
- Note the **Exercise button** works only one way, to make the loop less aggressive ("y"). However, whether the dynamic Exercise Mode (which strongly can adjust sensitivity, see [section 6.1.3.1](#)) is at work, cannot be recognized by the color of the Exercise button:

Dynamic Exercise Mode (dyn.EM) ?	bg target as in profile: grey	TT > profile target set (Yellow)
Exercise : grey	dyn.EM off	dyn.EM off
Exercise: yellow	dyn.EM <u>off (!) but</u> "on in waiting" = <u>after</u>	dyn.EM on

	an Automation sets a TT>profile target	
--	---	--

DEV idea: Should we make the **Y** to violet or orange (more aggressive) and keep the **y** yellow (less aggressive than profile)? Then beginners would easier recognize whether they go more or less aggressive ...

Color combinations	Lower aggressiveness		Higher aggressiveness
GGG		profile	
Gyy GyY	dynamic exercise mode (if TT > profile target)		EatingSoonTT (as TT < profile target: middle y inactive, GyY = GGY 2)
	<u>EXCEPTION</u> : Profile sensitivity is auto-adjusted in dynamic exercise mode, but this leaves the 1 st button grey (G)		
yGy-yGY- YGy-YGY	„traditional“ exercise mode (if <100% sens AND TT > profile target)	mixed cases 1)	boosted EatingSoonTT (if >100% sens AND TT < profile target)
GyG		GyG = GGG 2)	
yGG YGG	Loop adjusted for period of lower insulin need (e.g. 80% for night after exercise)		Loop adjusted for period of increased insulin need (e.g. 120% during sickness)
	The set %profile might be further adjusted by Activity Monitor or by Autosens		
GGy GGY	Loop running with temp elevated target e.g. for enhanced safety against going low (HypoTT)		EatingSoonTT (if TT < profile target)
yyG YyG	same as yGG	y inactive 2)	same as YGG
yyy - Yyy- yyY- YyY	dynamic exercise mode further modulated by <100% profile	mixed cases 3) yyY = yGY, 2)	Boosted EatingSoonTT (if >100% sens AND TT < profile target)(YyY = YGY, 2)

1) **yGY** would be the combination of setting a milder % profile, yet orienting the loop to a lower target. Usually we do not use this combination, but it would be OK for an instance where we want to start the next meal at a low target, but we want our loop to be careful and not rush getting there.

YGy would be the combination of setting a more aggressive % profile, yet orienting the loop to a higher target. Usually we do not use this combination, but it would be OK if we believe we are in a state of enhanced insulin need, and want to get to a set elevated TT (which could be what

we want to go max down to to have room for exercise, or just to be sure not to go into a hypoglycemia if we misjudged the symptoms about enhanced insulin need.

2) GyG is the same as GGG, differing only *if* (e.g. an Automation) sets a $TT > \text{profile target}$. The GyG setting *only then* provides softer loop response, due to the exercise mode kicking in.

Generally, a yellow middle button y works really only in the combination ..yy. Any combination ending yG or yY will work the same as if it were ending GG or GY. Although the yellow middle button does not make a difference in these cases, it might be “justified” by the loop showing its readiness to employ the exercise mode, whenever a $TT > \text{profile target}$ comes around, e.g. via an Automation.

3) yyY would essentially working like yGY (discussed under 1), because the middle button, whether y or G, makes no difference with an elevated TT.

Again, this the combination of setting a milder % profile, yet orienting the loop to a lower target. Usually we do not use this combination, but it would be OK for an instance where we want to start the next meal at a low target, but we want our loop to be careful and not rush getting there.

Yyy would be the combination of setting a more aggressive % profile, yet orienting the loop to a higher target. As opposed to the YGY case (discussed under 1), in Yyy the Exercise mode is active and strongly reduces aggressiveness. It is hard to imagine, why this combination this combination of settings could make sense; maybe if a “long waved” state of reduced insulin sensitivity meets a case where normally Gyy would be sought (see there).

5.3.3 Active Automations (or middleware in Trio, iAPS)

Active Automations can temporarily set different aggressiveness and, in AAPS, will automatically adjust to the color scheme discussed in the previous chapter.

It is important to be aware, which of your Automations (see listed in your AUTO tab, or via the top left burger menu, in AAPS) is **active to eventually “kick in”**.

The ones where you clicked “User Action” should also show as an **extra grey button** on the bottom of your AAPS screen (only in the suitable time-of-day bracket, if you assigned one).

Note that an **Automation might not be permitted** to change settings if still **another Automation is running**.

Always consider that. **Try to use short durations** in all your Automations! Note that usually you do not lose anything by that, because, if your trigger-Condition remained, the same Automation will repeatedly trigger.

755 For instance, you cannot switch from 130% profile to 110%. Either the 130% times out, or you
756 **need an extra “in-between” Automation that terminates** the 130% under described conditions
757 (example see in [Case study 6.2](#)).

758 This “design” is for a good reason: The assumption here is, that your 1st Automation (the
759 130% in the example) is designed well and runs for a reason. It should either “get finished”
760 when the job might be done (and kick in again, if not), or, in exceptional cases, it should be
761 consciously terminated by another well thought through 2nd Automation (describing the
762 conditions in which you would find that *another* Automation now is more important than
763 “finishing up” the one that was already running). That “in-between” Automation makes the
764 loop return to base profile, which is a signal *to all Automations*, to now check whether any
765 conditions exist, to activate a 3rd Automation. (An example is given in [Case study 6.2](#)).

766 Advice: 1) Do some “house cleaning”: Occasionally check which of your Automations might work
767 with shorter durations assigned. Reduce your long list of Automations, or at least de-activate those
768 that will not be needed.

769 In case you have many Automations and a very varied life-style, you could make it an evening
770 routine to have only those Automations activated that you might need next day.

771 2) Try to stay away from Automations that also aim at temp. modifying aggressiveness (especially
772 if triggered by bg level). Often, they will not kick in anyways. And generally, it is not a good idea, to
773 “double up” sub-algorithms for tweaking loop behaviors (“loop inside a loop”).(see also at end of
774 [section 5.1.4](#))

775

776 5.3.4 FCL related indicator fields in the AAPS home screen

777

778 In extra data fields of the AAPS main screen you can always see (not change) the key
779 „aggressiveness“ parameters your loop currently operates with (see also home screen
780 example below):

781 • To recognize whether autoISF currently runs with **ISF adaptation** or not, please consult the
782 **“ai: %” indicator** below the Autosens% **on the AAPS home screen**.

783 • Details for every loop decision see result/debug section of the **SMB tab**.

784 • The AAPS home screen additionally shows, above the deltas, the current **acceleration**

785 Having a look at that can be valuable. For instance, when glucose is relatively low and still
786 falling, a positive (and getting more positive) acceleration indicates that bg will swing back
787 up, rather than crash low. This will give info about necessary snack size, and hence help
788 avoid both, unnecessary calories, and going on a bg roller coaster.

789 5.3.5 Overall home screen: (do a new picture

790

Overall home screen:



<- Cockpit: yellow fields=>temp. modulated sens.

here additionally: acceleration-factor

Having a look at that can be valuable. For instance, when glucose is relatively low and still falling, a positive (and getting more positive) acceleration indicates that bg will swing back up, rather than crash low. This will give info about necessary snack size, and hence help avoid both, unnecessary calories, and going on a bg roller coaster.

<- Autosense status

<- Factor resulting from autoISF

<- buttons „bolus“ „carbs“ etc. eliminated

791

792 No more buttons „Insulin“, „Calculator“ etc at bottom of AAPS home screen

793

794 These buttons are not useful any longer in FCL, and should normally be removed (by going into
795 /preferences/overview/buttons and forcing them OFF).

796 The reason why we do this: It really is important to let the loop loop, and not interfere more than
797 absolutely needed. Any bolus the user gives will sure distort the bg curve, on which autoISF,
798 especially when aggressively tuned for FCL, builds a lot of its decisions!

799

800 Note: Eventual attractiveness of doing occasional pre-boli will be discussed in [section 7](#)

801

802 5.3.6 Extra Graphs for autoISF

803 The extra graphs available in AAPS below the main glucose chart, recently received an upgrade to include

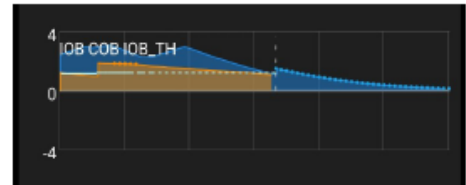
804 autoISF info: [\(English translation and more info to follow\)](#)

2. Anzeigen der ISF Faktoren – die werden als Linienplot in den kleineren Charts unterhalb gezeigt. Mit viel Arbeit habe ich es geschafft, die einzelnen Faktoren innerhalb des gleichen Charts alle mit der gleichen Skalierung innerhalb des selben Charts zu zeigen. Sie mit anderen Größen im gleichen Chart zu mixen, macht sowieso keinen Sinn. Weil final_ISF sowieso und acce_ISF meist viel größer sind, zeige ich die meist alleine sowie bei Bedarf die restlichen in einem anderen Chart.



In machen Fällen erklärt sich das Endergebnis nicht so leicht aus den Linien allein, nämlich wenn Exercise Mode oder ähnliche Effekte den Sensitivity Ratio als Basis schon modifizieren. Evtl. müsste der noch hinzugefügt werden. Das schon vorhandene Angebot SENS ist ja nur der Wert aus Autosens. Das hilft hier nicht.

3. Anzeigen des IOB-TH - der wird als Linie in den kleinen Charts angeboten. Es ist der gerade effektive iobTH, also User Setting plus Modifikationen aus z.B. Exercise Mode. Sinnvollerweise sollte er zusammen mit IOB gezeigt werden und deshalb sind auch ABS, IOB und IOB_TH alle untereinander gleich skaliert. Die Linie von IOB_TH ist gestrichelt, was aber manchmal trotzdem als volle Linie erscheint. Die Anzeige von COB kann man noch dazunehmen, denn sie wurde auch schon vorher passend zu IOB skaliert.



Ich glaube für den täglichen Gebrauch ist die Abbildung von IOB mit IOB_TH der wichtigste Zugewinn. Bei mir ist der Wert oft zu niedrig und das wird mir erst zu spät bewusst.

805

806



5.3.7 Info given every 5 minutes in the SMB tab

(If **tab** not shown -> Config.builder/APS: tick box on the right)

When clicking on the SMB tab, you see how your standard and temporary settings, as well as the latest bg and iob status, influenced the last decision of your FCL.

Example 1: A 80% temp. profile modulates 60% iobTH to 80% of 60% = 48%:



The profile ISF of 41 mg/dl/U got modified by the set 80% temp. profile to $41/0.8 = 51.3$ mg/dl/U, called “ISF unchanged” (before “start autoISF”).

autoISF applies the 4 sub-categories (acce, bg, pp and dura_ISF), and depending on the bg curve form suggests various ISF changes.

The final resulting factor “sens” (see flowcharts in [section 3](#)) is 1.11 (in our case, driven by bgAccel_ISF). This changes the 51.3 “unchanged” ISF to $51.3/1.11 = 42.4$ mg/dl/U

Further down in the SMB tab, you can see how this ISF is applied to define the SMB size to be given, and whether any limitations – notably by autoISFmax, max possible SMB size, or maxIOB – cut the amount.

```

≡ COMBOV2 SMB AUTO NSI :
Last run : 3/30/24 22:18

Result

Script debug : d:
  Activity monitor disabled
  inactivity detection: sleeping
  hours
  Autosens ratio: 1;
  Basal unchanged: 0.48;
  ISF unchanged: 51.3
  CR: 10

-----
start autoISF 3.0.1

-----
User setting iobTH=60%
modulated to 48% or 4.8U
due to profile % and/or
exercise/activity mode
SMB enabled; current target
90 is even number
Loop allows maximum power
acce_ISF adaptation is 1.11
bg_ISF adaptation is 1
pp_ISF adaptation is 1.09
dura_ISF adaptation is 1.03
because ISF 51.3 did not do it
for 15 m
final ISF factor is 1.11

-----
end autoISF

-----
currenttemp: 0.34
lastTempAge: 0 m
tempModulus: 28 m
  
```

Message	Condition	What does it affect?
Loop allows maximum power	even target < 100	increase in bg limited to 30%, otherwise no SMB; actual SMB delivery ratio is max of fixed smb_delivery_ratio and linearly growing ratio
Loop allows medium power	even target >= 100	increase in bg limited to 20%, the AAPS default, otherwise no SMB; actual SMB delivery ratio is either fixed smb_delivery_ratio or linearly growing ratio
Loop allows minimal power	odd target	no SMB, only TBR available for action
Loop power level temporarily capped	IOB > effective iobTH	Last SMB capped to stay below iob threshold + 30% overrun; user defined iobTH potentially modulated by exercise mode, activity monitor and profile percent
Loop allows AAPS power level	no even/odd target option active	SMB enabled/disabled according to standard AAPS rules and settings; no iobTH threshold is active

Note that in the SMB tab you can only capture and analyze 1 loop decision.

(Occasionally make screenshots ~ every 5 minutes in the critical stages, for analysis of a meal)

833 5.3.8 Info about last 15 autoISF decisions

834
835 Refer to [section 11.2.2](#) for an option that enables extended analysis of the on-going ISF modul-
836 ations from autoISF. (To do this on your **Android** loop phone requires QPython and a logfile
837 emulator).

838

839 You get tables like this ...

17:05

...

840
841 ... which gives you a quick orientation about recent loop decisions, and relative contributions of the
842 various autoISF contributors (ace, pp, bg, dura).

843

844 A similar table is available also on the **iPhone** for Trio and iAPS users of their autoISF variants.

845

846 5.3.9 SMB tab info when operating in 1-minute mode with Libre3

847

848 [Users: Please supply text and screenshot](#)

849

850

851

852 Summary: Your personal FCL cockpit for maneuvering through disturbances

853

854 A lot of avenues were shown that could help you or your loop maneuver through a variety of
855 “disturbances”. You should not have to try out many of them. Find a way to narrow it down to what
856 really helps in **your** everyday T1D management.

857

858 Try to **keep things as simple and clear as possible**.

859

860 Especially, do not pre-maturely rush into setting up Automations as quick over-patches for what
861 you may not like to see. Limit the number of Automations, and further limit which ones from your
862 list are every-day active (vs. switched off, and only ticked active for special days).

863 See also at end of [section 5.1.4](#)